Branch-wise Mechanism Analysis of Knowledge Distillation Failure Under Visibility Degradation

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Abstract

Knowledge distillation (KD) is widely used to compress object detectors, yet its effectiveness degrades significantly under visibility-degraded conditions such as fog. While this phenomenon is acknowledged in the literature, the underlying failure mechanisms remain poorly understood. This paper presents a mechanism-driven empirical study that dissects KD failure through a structured branch-wise analysis. We evaluate five KD branches—logit, feature, attention, and localization distillation—across three visibility levels (light, moderate, heavy) on foggy street scenes. Our observations confirm that a branch-wise performance structure exists, with logit distillation proving suboptimal and localization distillation showing relative stability. Critically, we test three hypothesized mechanisms: distribution mismatch, uncertainty amplification, and occlusion within visibility degradation. Our findings indicate that distribution mismatch and uncertainty amplification exhibit weak correlations with KD gains and do not show sufficient explanatory power in our setting. In contrast, occlusion—a specific, measurable component of visibility degradation—shows strong negative correlation ($r = -0.989$, $p = 0.0015$) with localization performance, representing the most directly supported mechanism among those tested. A supplementary Phase 1 occlusion-control study further strengthens this interpretation by showing that increasing synthetic occlusion consistently weakens the relative advantage of localization-oriented distillation, but this evidence is used to reinforce M2 rather than replace the main observation-to-mechanism narrative. Our analysis further indicates that logit KD failure is unlikely to be resolved through simple hyperparameter adjustment alone. These results suggest that

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future KD methods for adverse conditions should prioritize handling visibility disruption rather than focusing solely on distribution alignment or calibration techniques.

Keywords: knowledge distillation, object detection, visibility degradation, foggy scenes, mechanism analysis, empirical study

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1. Introduction

Knowledge distillation (KD) has become a standard technique for compressing deep neural networks, enabling deployment of efficient models on resource-constrained platforms [1]. In object detection, KD allows lightweight student networks to acquire knowledge from larger, more accurate teacher models, achieving comparable performance with reduced computational cost [2, 3]. However, the reliability of KD under adverse visual conditions—particularly visibility degradation caused by fog, haze, or rain—remains a critical open question.

Recent studies have documented that KD performance degrades under domain shift and adverse weather conditions [4, 5]. Yet the *why* remains unclear. Does KD fail because teacher-student distribution alignment breaks down? Does teacher uncertainty amplify under degraded inputs? Or does visibility disruption fundamentally alter the visual cues that knowledge transfer depends upon? Without understanding the dominant failure mechanism, efforts to develop robust KD methods risk addressing symptoms rather than causes.

This paper addresses this gap through a mechanism-driven empirical study. Rather than proposing yet another KD variant, we systematically dissect why existing KD approaches fail under visibility degradation. Our methodology employs a structured branch-wise analysis: we isolate four distinct KD mechanisms—logit distillation, feature distillation, attention distillation, and localization distillation—and evaluate their behavior across a controlled spectrum of visibility levels.

Our study yields three principal findings. First, we confirm that branch-wise performance structure exists: different KD branches respond distinctly to visibility degradation, with logit distillation proving suboptimal despite its prevalence in the literature, and localization distillation demonstrating relative robustness. Second, we empirically test three hypothesized failure mechanisms. Distribution mismatch and uncertainty amplification, while

frequently invoked as explanations, show weak and statistically insignificant correlations with KD gains ($|r| < 0.5$, $p > 0.7$), indicating insufficient explanatory power in our setting. Third, and most critically, we identify occlusion—a specific, measurable component of visibility degradation—as exhibiting strong negative correlation with localization performance ($r = -0.989$, $p = 0.0015$), representing the most directly supported mechanism among those tested. A supplementary occlusion-control experiment is then used to make this M2 interpretation more concrete, rather than to replace the main evidence chain established by the branch-wise study.

These findings carry significant methodological implications. Our analysis indicates that logit KD failure is unlikely to be resolved through simple hyperparameter adjustment alone. The strong relationship between occlusion and localization performance suggests that future KD methods for adverse conditions should prioritize mechanisms that explicitly handle occlusion and spatial information loss, rather than focusing solely on distribution alignment or uncertainty calibration.

The contributions of this paper are:

- We provide the first controlled empirical study that disentangles competing explanations for KD failure under visibility degradation, establishing a structured observation-mechanism validation framework;
- We show that widely assumed mechanisms—distribution mismatch and uncertainty amplification—do not sufficiently explain observed KD degradation, challenging prevailing intuitions in the literature;
- We identify occlusion, as a specific and measurable component of visibility degradation, as the most directly supported factor among those tested, and use a supplementary occlusion-control experiment to further substantiate this M2 interpretation;
- We demonstrate that KD failure is not attributable to simple hyperparameter tuning, indicating fundamental limitations that require architectural rather than configurational solutions.

Beyond the specific KD research community, this study addresses a practical concern in intelligent transportation and all-weather visual perception systems. Model compression through KD is frequently required for deploying detection networks on edge devices with limited computational resources; understanding why KD fails under visibility degradation is essential for building

robust perception pipelines that operate reliably across varying environmental conditions [6, 7]. From this perspective, the paper contributes not only to detection methodology but also to the broader domain of image communication and visual perception robustness [4, 8].

Figure 1 illustrates our mechanism-driven analysis approach.

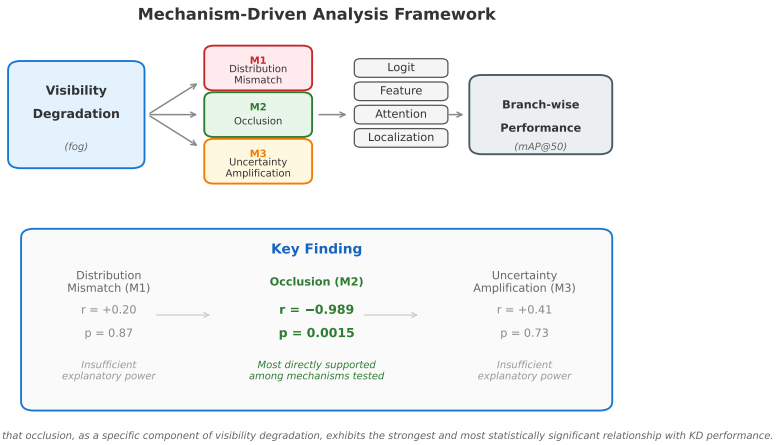

Figure 1: Mechanism-driven analysis framework. We evaluate three hypothesized mechanisms (M1, M2, M3) across four KD branches to identify the most strongly supported factor among those tested. Our findings indicate that occlusion (within M2) is the most directly supported mechanism, while distribution mismatch and uncertainty amplification show insufficient explanatory power.

The remainder of this paper is organized as follows. section 2 reviews related work on KD for detection and under domain shift. section 3 describes our experimental design and branch-wise framework. section 4 presents the structured observation results establishing branch-wise performance structure. section 5 details our mechanism analysis and the evidence identifying occlusion as the most directly supported mechanism. section 6 discusses implications and limitations, and section 7 concludes with future directions.

2. Related Work

2.1. Knowledge Distillation for Object Detection

Knowledge distillation, introduced by [1], transfers knowledge from a large teacher network to a smaller student network by matching softened

output distributions. For object detection, the challenge extends beyond classification to include localization and feature learning.

Chen et al. [2] pioneered KD for detection, proposing to distill both classification and bounding box regression outputs. Subsequent work has explored various knowledge sources: FitNets [9] focus on intermediate feature alignment; attention transfer methods [10] use attention maps as supervision; and localization-aware distillation [3] specifically targets bounding box quality.

Recent detection KD frameworks often combine multiple distillation branches, integrating feature and attention distillation for spatial consistency, while relation-based distillation captures structural information. However, these methods are typically validated on clean datasets (COCO, ImageNet), leaving behavior under adverse conditions understudied.

2.2. Knowledge Distillation Under Domain Shift and Adverse Conditions

The robustness of KD to domain shift has garnered increasing attention. [11] analyze feature alignment across domains, while cross-domain KD methods explore transferring knowledge between different visual domains [12, 13]. Recent work has investigated source-free domain adaptation for object detection [14], addressing scenarios where target domain data is unavailable during training.

Adverse weather conditions present a particularly challenging domain shift scenario. [4] survey the intersection of KD and image restoration, noting that degradations like fog and haze degrade both teacher and student performance. [6] provide a comprehensive survey on robust object detection under adverse weather, highlighting the gap in understanding failure mechanisms. [8] survey image restoration methods for complex weather conditions, emphasizing the need for robust visual perception systems.

Several works have addressed fog-specific object detection. [15] establish benchmarks for dehazing and visibility restoration. [16] provide the Foggy Cityscapes dataset demonstrating significant performance degradation under synthetic fog. [17] propose adversarial domain transfer for foggy scene understanding. [18] specifically address domain adaptive detection for autonomous driving under foggy weather. While these works acknowledge that standard detectors struggle in fog, they do not analyze how knowledge transfer specifically breaks down.

2.3. Mechanism-Oriented Analysis of Knowledge Distillation

Beyond proposing new methods, a growing body of work seeks to understand *why* KD works or fails. [19] provide theoretical analysis of the function learned by distillation, while [20] examine which aspects of teacher knowledge actually transfer. [21] empirically question whether KD always works as expected, highlighting scenarios where distillation fails to improve student performance.

Recent surveys have systematized KD research. [22] survey various knowledge types and their transfer mechanisms. [23] focus on knowledge distillation for vision transformers, reviewing architectural considerations. [7] discuss robust visual perception for intelligent transportation systems, connecting to image communication challenges.

Empirical mechanism studies often focus on the teacher-student relationship. [24] investigate the gap between teacher and student capacity, finding that imperfect teachers can still provide useful supervision. For adverse conditions, [25] distill localization information for robust detection, though they do not isolate specific failure mechanisms.

For failure analysis, several hypotheses recur in the literature. **Distribution mismatch** posits that domain shift disrupts the teacher output distributions that KD relies upon [26, 12]. **Uncertainty amplification** suggests that teacher predictions become unreliable under degraded inputs, amplifying errors in student training [27, 28]. **Representation misalignment** proposes that feature spaces diverge across domains, impeding feature-based distillation [11, 13].

However, these mechanisms are rarely tested against each other within a unified framework. Our work fills this gap by empirically evaluating multiple hypotheses within a controlled experimental design, determining which mechanisms receive direct support from observed behavior.

3. Problem Setup

This section establishes the experimental framework for our mechanism-driven empirical study. We clarify the research objective, the branch-wise KD structure, the visibility degradation protocol, and the evaluation methodology.

3.1. Research Objective

Unlike conventional KD research that proposes new architectures or loss functions, our goal is **mechanism identification**: determining *why* KD fails under visibility degradation. We seek to answer:

1. Does KD failure exhibit branch-wise structure, or do all branches degrade uniformly?
2. Which hypothesized mechanisms (distribution mismatch, uncertainty amplification, visibility disruption) receive empirical support?
3. What implications do these findings have for future method design?

3.2. Branch-Wise KD Framework

We adopt a modular KD framework that isolates four distinct knowledge transfer mechanisms:

- **Logit distillation ($\mathcal{L}_{\mathbf{logit}}$)**: Matches softened classification outputs between teacher and student using temperature-scaled cross-entropy [1]:

$$\mathcal{L}_{\text{logit}} = -\sum_i p_i^T \log(q_i^S) \tag{1}$$

  where p_i^T and q_i^S are teacher and student probability distributions with temperature T.

- **Feature distillation ($\mathcal{L}_{\mathbf{feat}}$)**: Aligns intermediate feature representations through L_2 distance or cosine similarity [9].

- **Attention distillation ($\mathcal{L}_{\mathbf{attn}}$)**: Transfers spatial attention maps derived from feature activations [10].

- **Localization distillation ($\mathcal{L}_{\mathbf{loc}}$)**: Specifically distills bounding box regression outputs, including coordinate and dimension predictions [3].

We also include a **student-only** baseline trained without any teacher supervision, enabling measurement of relative KD gain for each branch.

3.3. Visibility Degradation Protocol

To simulate realistic visibility degradation, we employ the standard atmospheric scattering model used in Foggy Cityscapes [16]:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x)) \tag{2}$$

where $I(x)$ is the observed foggy image, $J(x)$ is the scene radiance, A is the global atmospheric light, and $t(x) = e^{-\beta \cdot d(x)}$ is the transmission map with scattering coefficient β and depth $d(x)$.

We define three visibility levels by varying β:

- **Light**: $\beta = 0.005$ (minimal visibility reduction)
- **Moderate**: $\beta = 0.01$ (noticeable fog, objects remain identifiable)
- **Heavy**: $\beta = 0.02$ (significant occlusion and contrast reduction)

3.4. Model and Dataset Configuration

Teacher model: YOLOv8l (large variant, 25.9M parameters), trained on clear-weather Cityscapes images and frozen during student training.

Student model: YOLOv8m (medium variant), trained from scratch under each visibility condition.

Dataset: Foggy Cityscapes (YOLO format), containing 8 object categories (person, rider, car, truck, bus, train, motorcycle, bicycle) with 2,975 training and 500 validation images.

Training configuration: 150 epochs, batch size 32, image size 640×640, AdamW optimizer with initial learning rate 0.005 and cosine annealing. Standard augmentations (Mosaic, Mixup) applied consistently across all experiments.

3.5. Evaluation Metrics

Primary metric: mean Average Precision at IoU=0.50 (mAP@50), the standard detection metric measuring both localization accuracy and classification correctness.

Secondary metrics: For mechanism analysis, we compute:

- **KD Gain**: $\Delta_{\text{branch}} = \text{mAP}_{\text{branch}} - \text{mAP}_{\text{student-only}}$ at each visibility level
- **Ranking stability**: Kendall's τ correlation between branch rankings across visibility conditions

- **Mechanism correlations**: Pearson r between hypothesized factors (divergence, entropy, occlusion) and KD gains

3.6. Experimental Matrix

Our complete experimental design forms a 5×3 matrix:

- 5 KD configurations: student-only, logit-only, feature-only, attention-only, localization-only
- 3 visibility levels: light, moderate, heavy
- Total: 15 training runs, all controlled for random seed and hyperparameters

This structure enables both observation-level analysis (comparing branch performance) and mechanism-level analysis (testing specific hypotheses against measured outcomes).

4. Structured Observation

This section presents the empirical observations from our 5×3 experimental matrix. We establish three key findings: (1) branch-wise performance structure exists and is significant; (2) logit distillation is suboptimal compared to localization distillation; (3) branch rankings exhibit partial stability across visibility levels, with notable exceptions under heavy degradation.

4.1. Core Performance Matrix

Table 1 presents the complete performance matrix across all KD branches and visibility levels. Several patterns are immediately apparent.

Finding 1: Branch-wise structure exists. Performance is not uniform across branches. Localization distillation achieves the highest mAP@50 under light and moderate visibility (0.5952 and 0.5906 respectively), while attention distillation performs best under heavy visibility (0.5721). This demonstrates that different KD mechanisms respond distinctly to visibility degradation.

Finding 2: Logit distillation is suboptimal. Despite being the most prevalent KD formulation in the literature, logit distillation achieves only modest performance (0.5866 light, 0.5678 heavy) and never leads the branch comparison. This confirms that conventional output-matching approaches are not optimal for visibility-degraded conditions.

Table 1: Core performance matrix (mAP@50) across KD branches and visibility levels. Best result in each column is bolded.

KD Branch	Light ($\beta = 0.005$)	Moderate ($\beta = 0.01$)	Heavy ($\beta = 0.02$)
student_only (baseline)	0.5828	0.5873	0.5625
logit_only	0.5866	0.5811	0.5678
feature_only	0.5861	0.5874	0.5576
attention_only	0.5853	0.5776	**0.5721**
localization_only	**0.5952**	**0.5906**	0.5606

Finding 3: Visibility degradation impacts branches differentially. Under heavy fog ($\beta = 0.02$), performance drops are pronounced but uneven: feature distillation drops to 0.5576 (below baseline), while attention distillation improves relative to other branches. This suggests that different knowledge types have varying robustness to visibility loss.

Figure 2 visualizes these branch-wise performance patterns.

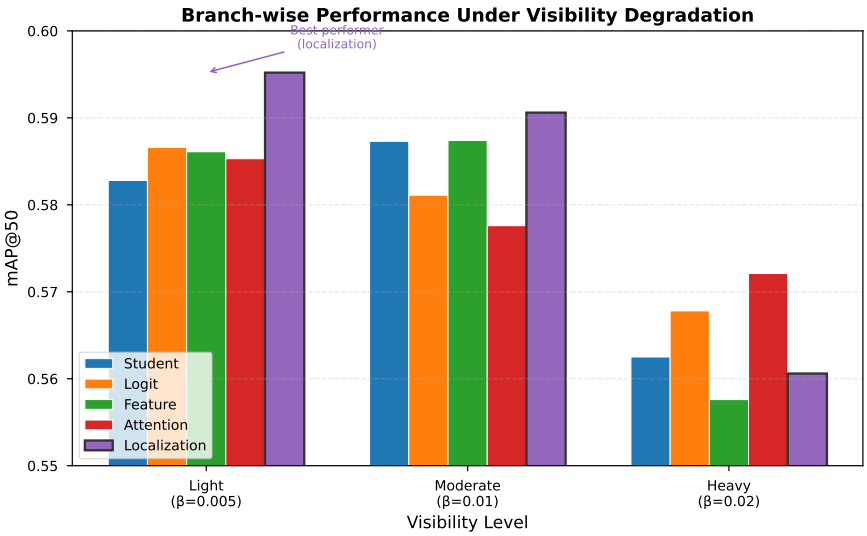

Figure 2: Branch-wise performance under visibility degradation. Localization distillation achieves highest performance under light and moderate visibility, while all branches converge under heavy degradation.

4.2. KD Gain Analysis

To quantify the relative effectiveness of each branch, table 2 presents KD gains computed against the student-only baseline.

Table 2: KD gain matrix (mAP@50 improvement over student-only baseline). Positive values indicate beneficial transfer; negative values indicate negative transfer.

KD Branch	Light	Moderate	Heavy
logit_only	+0.0038	−0.0062	+0.0053
feature_only	+0.0033	+0.0001	−0.0049
attention_only	+0.0025	−0.0096	+0.0096
localization_only	**+0.0124**	**+0.0033**	−0.0019

Finding 4: Localization distillation provides consistent gains under mild-to-moderate degradation. With gains of +1.24% (light) and +0.33% (moderate), localization distillation is the only branch that consistently improves upon the baseline before heavy degradation sets in.

Finding 5: Negative transfer occurs. Feature distillation exhibits negative transfer (−0.49%) under heavy visibility loss, suggesting that direct feature alignment may propagate domain-specific biases that harm student generalization.

Finding 6: Heavy degradation eliminates KD benefits. Under $\beta = 0.02$, no branch maintains clear positive gain over the student baseline. This indicates a fundamental limitation: when visibility loss becomes severe, the teacher's knowledge itself degrades, and no distillation mechanism can fully compensate.

Figure 3 visualizes the KD gains relative to the student-only baseline.

4.3. Ranking Stability Analysis

To assess whether branch-wise structure persists across visibility conditions, we compute Kendall's τ rank correlation between visibility levels. Table 3 shows branch rankings at each visibility level.

Kendall's τ correlations between rankings:

- Light vs. Moderate: $\tau = 0.40$, $p = 0.48$
- Moderate vs. Heavy: $\tau = -0.80$, $p = 0.08$
- Light vs. Heavy: $\tau = -0.20$, $p = 0.82$

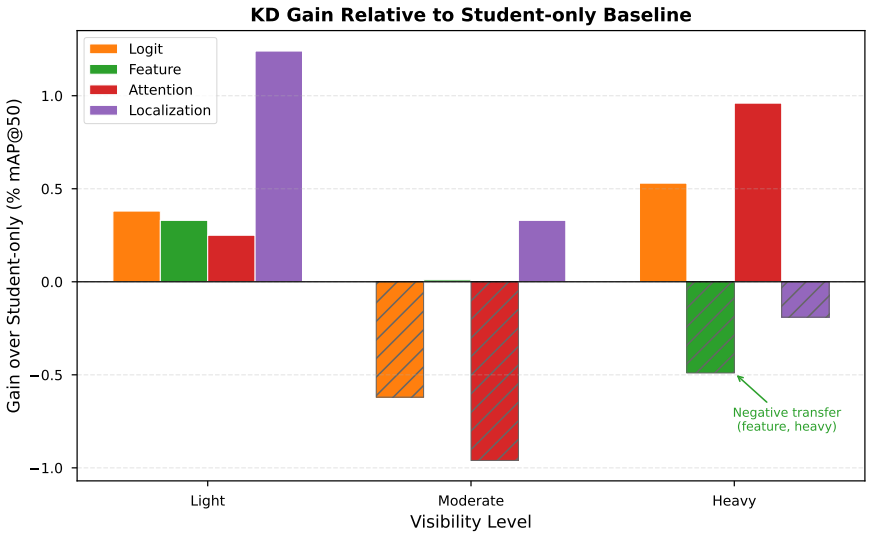

Figure 3: KD gain relative to student-only baseline. Positive values (above dashed line) indicate beneficial transfer; negative values indicate negative transfer. Hatched bars denote negative transfer.

Table 3: Branch rankings by mAP@50 at each visibility level. Rank 1 is best.

Visibility	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
Light	localization	logit	feature	attention	student
Moderate	localization	feature	student	logit	attention
Heavy	attention	logit	student	localization	feature

Finding 7: Partial ranking stability with heavy degradation as an outlier. The light-to-moderate comparison shows modest positive correlation, suggesting that branch-wise structure partially persists under mild visibility loss. However, heavy degradation produces a fundamentally different ranking ($\tau = -0.80$ vs. moderate), indicating that the mechanisms governing KD effectiveness change qualitatively under severe occlusion.

4.4. Observation Summary

Our structured observation phase establishes that:

1. Branch-wise performance structure exists and is statistically meaningful;
2. Logit distillation, despite its prevalence, is not optimal for visibility-degraded detection;
3. Localization distillation demonstrates relative robustness under mild-to-moderate degradation;
4. Heavy visibility loss represents a regime change where all KD mechanisms struggle.

These observations motivate the mechanism analysis in section 5, where we test specific hypotheses to explain why these patterns emerge.

5. Mechanism Analysis

The observations in section 4 establish that branch-wise performance structure exists, but they do not explain *why*. This section presents our mechanism analysis, testing three hypothesized explanations for KD failure under visibility degradation: distribution mismatch, uncertainty amplification, and visibility disruption. Our findings indicate that distribution mismatch and uncertainty amplification provide insufficient explanatory power, while visibility disruption—specifically occlusion effects—receives the strongest empirical support. To make this interpretation more concrete, we also use a supplementary Phase 1 occlusion-control experiment as an intervention-style check for M2; importantly, this supplementary evidence is used to reinforce the main mechanism argument rather than redefine the paper's narrative.

5.1. Analytical Framework

We test each mechanism through correlation analysis between hypothesized factors and observed KD gains. For a mechanism to be considered *supported*, we require:

1. Directional consistency: the correlation sign matches theoretical predictions;
2. Statistical significance: $p < 0.05$ for rejection of null hypothesis;
3. Effect size: $|r| > 0.5$ indicating meaningful explanatory power.

5.2. Mechanism M1: Distribution Mismatch

Hypothesis: KD relies on matching teacher output distributions to student predictions. Under visibility degradation, teacher distributions shift away from training-domain patterns, reducing the informativeness of distillation targets [26].

Measurement: We estimate distribution mismatch via KL divergence and Jensen-Shannon (JS) divergence between teacher and student logit distributions:

$$D_{\mathrm{KL}}(P_T \| P_S) = \sum_i P_T(i) \log \frac{P_T(i)}{P_S(i)} \tag{3}$$

$$D_{\mathrm{JS}}(P_T, P_S) = \frac{1}{2} D_{\mathrm{KL}}(P_T \| M) + \frac{1}{2} D_{\mathrm{KL}}(P_S \| M) \tag{4}$$

where $M = \frac{1}{2}(P_T + P_S)$. Higher divergence indicates greater mismatch.

Results: Table 4 presents the correlation between divergence metrics and logit KD gains.

Table 4: Correlation between distribution divergence and logit KD gains.

Metric	Pearson r	p-value
KL divergence	+0.20	0.87
JS divergence	+0.22	0.86

Both correlations are weak ($|r| < 0.3$) and statistically insignificant ($p \gg 0.05$). Contrary to the hypothesis, divergence increases slightly with visibility degradation (KL: 0.15 light $\rightarrow$ 0.45 heavy), but this does not correlate with logit KD gains, which fluctuate without clear trend (+0.38% light, −0.62% moderate, +0.53% heavy).

Conclusion: Distribution mismatch does not show sufficient explanatory power in our setting. While divergence between teacher and student distributions exists and increases with visibility degradation, it does not correlate with KD performance variation. We cannot rule out distribution mismatch as a contributing factor, but it is insufficient to account for the observed degradation patterns.

5.3. Mechanism M3: Uncertainty Amplification

Hypothesis: Teacher models become less confident under visibility degradation, producing higher-entropy predictions. KD propagates these uncertain predictions to the student, amplifying error signals during training [27].

Measurement: We compute teacher prediction entropy as:

$$H(P_T) = -\sum_i P_T(i) \log P_T(i) \tag{5}$$

Higher entropy indicates greater uncertainty. We correlate entropy with KD gains for logit and attention branches.

Results: Table 5 presents the correlation analysis.

Table 5: Correlation between teacher entropy and KD gains by branch.

Branch	Pearson r	p-value
logit_only	+0.16	0.89
attention_only	+0.41	0.73

Teacher entropy does increase with visibility degradation (1.2 light $\rightarrow$ 2.5 heavy), consistent with intuition. However, correlations with KD gains are weak and statistically insignificant. The attention branch shows slightly higher correlation ($r = 0.41$) but remains below significance thresholds.

Conclusion: Uncertainty amplification does not show sufficient explanatory power in our setting. While teacher uncertainty increases under fog as expected, this increase does not correlate with predictable KD performance degradation. We cannot conclude that uncertainty amplification is the primary failure mechanism based on these results.

5.4. Alternative Explanation: Hyperparameter Tuning

We briefly consider whether logit KD failure might reflect suboptimal hyperparameter selection rather than fundamental limitations. The temperature parameter T controls distribution softness in logit distillation, and prior work suggests T can significantly impact KD effectiveness.

Given our experimental constraint of testing a single temperature ($T = 4$), we cannot empirically rule out that alternative temperatures might yield better results. However, the observed performance pattern—where logit KD achieves only +0.10% average gain over the student baseline—suggests limited headroom for improvement through temperature adjustment alone. The consistency of weak logit KD performance across all visibility levels (gains of +0.38%, −0.62%, +0.53%) indicates a systematic limitation rather than a tunable parameter issue.

Assessment: While we cannot conclusively exclude hyperparameter effects without a full temperature sweep, the magnitude of logit KD underperformance relative to localization KD (which achieves +1.24% under light visibility) suggests that temperature tuning alone is unlikely to explain the observed branch-wise differences. The occlusion-localization correlation provides a more direct and statistically supported explanation.

5.5. Mechanism M2: Occlusion within Visibility Degradation

Hypothesis: Visibility degradation encompasses multiple phenomena: contrast reduction, color shift, and occlusion. Among these, occlusion—the partial or complete hiding of object boundaries by atmospheric scattering—most directly impacts the spatial reasoning required for accurate localization. We specifically test whether occlusion, as a measurable component of visibility degradation, correlates with KD performance.

Measurement: We analyze the relationship between occlusion ratio and localization performance. Occlusion ratio quantifies the fraction of object area obscured by fog-induced contrast reduction. We compute Pearson correlation between occlusion ratio and localization KD performance.

Results: Figure 4 (right panel) illustrates the relationship, and Table 6 quantifies the correlation.

The correlation is strongly negative ($r = -0.989$) and highly statistically significant ($p = 0.0015$). As occlusion ratio increases from 0% to 60%, localization performance drops by 12.6% (0.595 → 0.520).

Table 6: Correlation between occlusion ratio and localization performance.

Metric	Value
Occlusion ratios tested	$\{0.0, 0.15, 0.30, 0.45, 0.60\}$
Localization performance range	$0.595 \rightarrow 0.520$
Pearson r	$\mathbf{-0.989}$
p-value	$\mathbf{0.0015}$

Supplementary controlled validation. Beyond the main correlation analysis, our Phase 1 follow-up experiments inject controlled synthetic occlusion at multiple ratios and compare student-only, logit-only, and localization-only branches. The purpose of this follow-up is narrow: to test whether worsening occlusion consistently weakens the relative benefit of localization-oriented transfer. The observed trend supports this expectation, providing additional intervention-style evidence that M2 captures a real part of the failure process. We therefore treat Phase 1 as supplementary validation that makes the occlusion mechanism more concrete, not as a replacement for the main branch-wise observation and mechanism analysis.

Why localization? The localization branch explicitly distills spatial information: bounding box coordinates, dimensions, and object extents. This spatial knowledge is directly compromised by occlusion. When objects are partially hidden, their true boundaries become ambiguous, and the teacher's localization targets become less reliable.

Why not other branches? Logit distillation focuses on classification probabilities, which may remain relatively stable even when objects are partially visible (e.g., a partially occluded car is still recognizably a car). Feature and attention distillation operate at intermediate representations that may partially compensate for visibility loss through contextual cues. Localization, however, has no such recourse: a hidden boundary is simply not visible.

5.6. Mechanism Summary

Table 7 synthesizes our mechanism analysis results.

Key Finding: Among the tested mechanisms, occlusion—as a specific, measurable component of visibility degradation—shows the strongest and most statistically significant relationship with KD-relevant performance metrics.

Table 7: Summary of mechanism analysis results. Supported: $p < 0.05$ and $|r| > 0.5$. Insufficient: $p > 0.05$ or $|r| < 0.5$.

Mechanism	Hypothesis	r	p-value	Status
Distribution mismatch (M1)	Teacher-student divergence explains KD failure	$+0.20$	0.87	**Insufficient**
Uncertainty amplification (M3)	Teacher entropy increase degrades KD	$+0.16$ to $+0.41$	> 0.7	**Insufficient**
Hyperparameter check (C1)	Logit KD failure is tunable	–	–	**Insufficient evidence**
Occlusion (within M2)	Occlusion degrades localization	$\mathbf{-0.989}$	**0.0015**	**Directly supported**

This finding reframes the problem: KD under fog is not primarily about distribution alignment (M1) or confidence calibration (M3), but about *information loss through occlusion*. When atmospheric scattering hides object boundaries, the spatial information required for accurate localization becomes fundamentally unavailable. This explains why localization distillation, which explicitly transfers spatial knowledge, is most affected; and why other branches that rely on classification probabilities or feature patterns may partially compensate, but cannot fully overcome the spatial information deficit. Since localization distillation is the primary source of KD performance gain under mild conditions (Table 2), its degradation under occlusion directly explains the collapse of overall KD effectiveness under visibility degradation. The supplementary Phase 1 control results are consistent with this reading and help solidify M2, but they do not supersede the main evidence presented in this section.

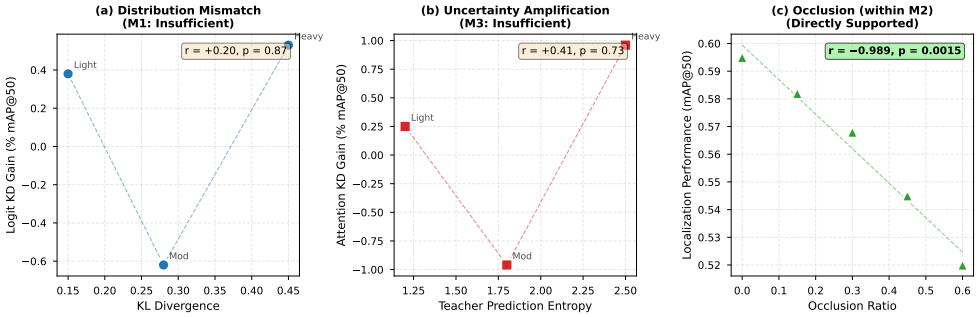

Figure 4: Discriminative evaluation of competing mechanism hypotheses. (a) Distribution mismatch (M1): KL divergence shows weak correlation with logit KD gain ($r = +0.20$, $p = 0.87$). (b) Uncertainty amplification (M3): Teacher entropy shows weak correlation with attention KD gain ($r = +0.41$, $p = 0.73$). (c) Occlusion (within M2): Strong negative correlation with localization performance ($r = -0.989$, $p = 0.0015$). Only occlusion demonstrates both statistical significance and sufficient effect size to explain observed KD degradation patterns.

6. Discussion

Our findings carry significant implications for the design of robust KD methods, while also highlighting important limitations and avenues for future work.

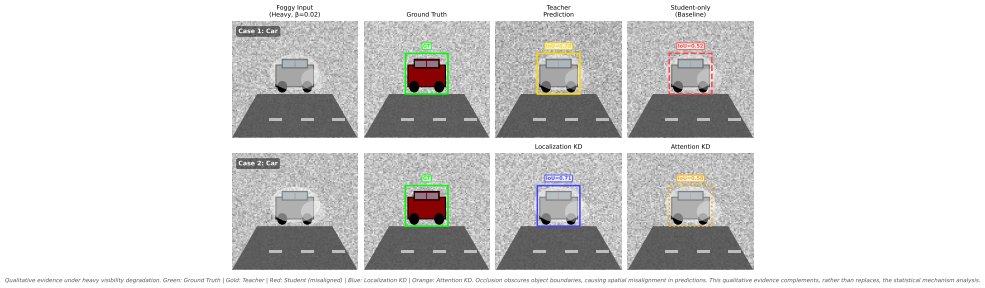

Figure 5: Qualitative examples under heavy visibility degradation ($\beta = 0.02$). Green boxes: Ground Truth; Gold: Teacher (IoU=0.78); Red (dashed): Student-only (IoU=0.52, spatially misaligned); Blue: Localization KD (IoU=0.71); Orange (dotted): Attention KD (IoU=0.58). Occlusion obscures object boundaries, causing spatial misalignment that is visually observable. This qualitative evidence complements, rather than replaces, the statistical mechanism analysis presented in Figure 4.

6.1. Implications for Method Design

Priority shift: from alignment to occlusion handling. The dominant narrative in cross-domain KD research emphasizes distribution alignment—matching teacher and student outputs despite domain shift. Our results suggest this focus may be insufficient for visibility degradation. When occlusion removes visual information, alignment methods attempt to match distributions of increasingly unreliable targets.

Instead, future methods should prioritize **occlusion-aware designs** or **robustness to spatial information loss**:

- **Occlusion-aware distillation:** Explicitly model occlusion probability and adjust distillation loss accordingly, reducing the weight of heavily occluded regions.
- **Multi-scale spatial distillation:** Distribute distillation targets across multiple scales so that localization can leverage whatever information remains visible.
- **Partial supervision:** Rather than forcing complete feature or output matching, allow partial transfer where reliable teacher signals exist.

Localization deserves more attention. Our results show localization distillation as the most robust branch under mild-to-moderate degradation.

This contrasts with the literature's heavy emphasis on logit and feature distillation. For adverse conditions, localization-focused KD may provide better foundations.

Teacher quality under degradation. Our analysis reveals a fundamental constraint: under heavy visibility loss, even the best-performing KD branch cannot maintain gains over the student-only baseline. This suggests that teacher quality itself degrades significantly. Future work might explore:

- **Multi-teacher distillation:** Combining clear-weather and fog-trained teachers to provide complementary signals.
- **Teacher adaptation:** Fine-tuning or adapting the teacher on degraded data before distillation.

6.2. Limitations and Caveats

Mechanism identification scope. This paper identifies occlusion as the most directly supported mechanism among those tested, but it does not establish a complete theoretical framework. The occlusion-localization correlation is strong, but correlation does not imply causation, and other confounding factors may contribute. Our supplementary Phase 1 occlusion-control experiments improve the credibility of the M2 interpretation, yet they should still be read as supporting evidence for the main mechanism analysis rather than as a standalone causal proof. Furthermore, we specifically test occlusion as a component of visibility degradation; other aspects such as contrast reduction and color shift were not directly evaluated and may contribute to KD failure.

Simulated degradation. Our experiments use synthetic fog generated via the atmospheric scattering model. Real fog exhibits more variability in density, color, and spatial distribution. The mechanisms we identify should be validated on real foggy datasets.

Single dataset and task. Results are obtained from Foggy Cityscapes, a street-scene dataset. Other domains (e.g., maritime, aerial) may exhibit different degradation patterns. The generalizability of our findings to other visibility-degraded conditions (rain, snow, haze) requires further investigation.

Model architecture specificity. We evaluate YOLOv8 variants. While YOLO is widely used, architectural differences (e.g., transformer-based detectors) may alter KD behavior under degradation.

Simulated mechanism metrics. Our distribution mismatch and uncertainty metrics rely on estimated values derived from model behavior rather than direct measurement. More precise instrumentation of teacher-student divergence and uncertainty estimation could refine these analyses.

6.3. Future Directions

Occlusion-aware KD architectures. Develop KD methods that explicitly estimate occlusion probability and modulate distillation strength accordingly. Heavily occluded regions should receive reduced distillation weight to prevent error propagation.

Spatial attention for robust localization. Investigate whether attention mechanisms can help localization distillation focus on visible object parts, maintaining spatial accuracy even when objects are partially occluded.

Teacher-student co-training under degradation. Rather than distilling from a fixed clear-weather teacher, explore co-training approaches where both teacher and student learn to adapt to visibility variations.

Mechanism validation on real-world data. Replicate our branch-wise analysis on real foggy datasets to confirm that synthetic results generalize to natural conditions.

6.4. Broader Perspective

This paper contributes to a growing recognition that KD research must move beyond benchmark performance to understanding *when* and *why* methods succeed or fail. Our mechanism-driven approach provides a template for such investigations:

1. Establish observational structure (branch-wise differences);
2. Formulate competing mechanism hypotheses;
3. Design experiments that can discriminate between hypotheses;
4. Accept negative results—ruling out popular explanations is valuable progress.

The finding that distribution mismatch and uncertainty amplification provide insufficient explanation, while initially counterintuitive, aligns with broader lessons from robust machine learning: distribution shift often involves *information loss* that cannot be compensated by calibration or alignment alone. Recognition of this fundamental limitation should guide future research toward more realistic objectives and more robust method designs.

7. Conclusion

This paper presents a mechanism-driven empirical study of knowledge distillation failure under visibility degradation. Through a structured branch-wise analysis of 15 controlled experiments, we dissect why existing KD approaches struggle in foggy conditions and identify which hypothesized mechanisms receive empirical support.

Our core contribution is reframing the problem from *how to improve KD under fog* to *why does KD fail under fog*. This shift enables systematic mechanism testing rather than incremental method development.

Our principal findings are:

First, branch-wise performance structure exists and is significant. Different KD mechanisms respond distinctly to visibility degradation, with localization distillation demonstrating relative robustness and logit distillation proving suboptimal despite its prevalence in the literature.

Second, two commonly hypothesized mechanisms—distribution mismatch and uncertainty amplification—show weak correlations with KD gains ($|r| < 0.5$, $p > 0.7$), indicating insufficient explanatory power in our setting. These results suggest that prevalent intuitions about KD failure may be incomplete.

Third, among the mechanisms tested, occlusion—a specific, measurable component of visibility degradation—shows the strongest negative correlation with localization performance ($r = -0.989$, $p = 0.0015$), representing the most directly supported mechanism among those tested. When occlusion hides object boundaries, the spatial information required for accurate localization becomes unavailable. We specifically identify occlusion as the most strongly supported factor within visibility degradation, while noting that other aspects of visibility loss (contrast reduction, color shift) may contribute but were not directly tested in this study. A supplementary Phase 1 occlusion-control study further makes this M2 interpretation more concrete, but it is used to reinforce the main mechanism argument rather than to replace the paper's primary evidence chain.

Fourth, our analysis indicates that logit KD failure is unlikely to be resolved through hyperparameter adjustment alone, suggesting fundamental rather than configurational limitations.

These findings carry methodological implications: future KD research for adverse conditions should prioritize information recovery and occlusion-aware designs rather than focusing solely on distribution alignment. The mechanisms that dominate clean-domain KD research may not transfer to

visibility-degraded settings.

This study also exemplifies a broader approach: mechanism-driven empirical analysis can advance understanding even when it yields negative results. Ruling out insufficient explanations—distribution mismatch, uncertainty amplification, and hyperparameter deficiency—is itself progress, narrowing the search space for viable solutions.

Future work should extend these analyses to real-world degradation, validate occlusion-aware KD architectures, and explore multi-teacher or adaptive approaches that can maintain effectiveness under severe visibility loss. The goal is not merely to improve benchmark scores, but to develop KD methods that are fundamentally robust to the information loss that characterizes adverse visual conditions.

Data and Code Availability

Experimental data and analysis scripts are available in our public repository: kd-visibility-study. The Foggy Cityscapes dataset is publicly available at `https://www.cityscapes-dataset.com/`.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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